

Product Requirement Document (PRD)

IntelliPlant AI

AI-Powered Industrial Knowledge Intelligence Platform

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1. Executive Summary

IntelliPlant AI is an Industrial Knowledge Intelligence Platform designed to unify fragmented industrial documents, operational records, maintenance history, engineering drawings, and compliance documents into a centralized AI-powered knowledge brain.

The platform enables engineers, maintenance teams, quality teams, and plant managers to access critical information through conversational AI, semantic search, knowledge graphs, and intelligent agents.

The objective is to reduce information retrieval time, preserve organizational knowledge, improve compliance, minimize downtime, and enhance operational decision-making.

2. Problem Statement

Industrial organizations operate with multiple disconnected systems containing:

- Engineering drawings
- Maintenance work orders
- SOP documents
- OEM manuals
- Incident reports
- Inspection records
- Email archives
- Compliance documents
- Quality records

Engineers spend significant time searching for information.

Consequences include:

- Increased downtime
- Knowledge loss
- Poor decision making
- Compliance failures
- Repeated incidents
- Operational inefficiency

The organization lacks a unified intelligence layer capable of transforming static documents into actionable knowledge.

3. Vision Statement

Create an AI-powered Industrial Knowledge Intelligence Platform that transforms fragmented documents into a continuously evolving operational brain accessible to every stakeholder across devices and departments.

4. Product Goals

Goal 1

Reduce information retrieval time from 30–60 minutes to less than 20 seconds.

Goal 2

Capture and preserve tribal knowledge before expert retirement.

Goal 3

Enable explainable AI-based engineering assistance.

Goal 4

Improve maintenance efficiency and root cause analysis.

Goal 5

Identify compliance gaps proactively.

Goal 6

Provide lessons learned intelligence across departments.

5. Business Objectives

Operational Excellence

Reduce unplanned downtime.

Knowledge Preservation

Prevent loss of expert knowledge.

Safety Enhancement

Reduce repeated incidents.

Compliance Management

Improve audit readiness.

Productivity Improvement

Increase engineer efficiency.

Digital Transformation

Convert documents into organizational intelligence.

6. Target Industries

Oil & Gas

Refineries

Pipelines

Petrochemical Plants

Manufacturing

Steel Plants

Automobile Plants

Process Industries

Energy

Power Plants

Renewable Energy Facilities

Transmission Systems

Mining

Coal Mines

Metal Processing Plants

Chemical Industries

Process Plants

Fertilizer Plants

EPC Organizations

Infrastructure Projects

Data Centers

Industrial Construction

7. User Personas

Maintenance Engineer

Responsibilities:

- Troubleshooting failures
- Reviewing maintenance history
- Root cause analysis

Pain Points:

- Searching manuals
- Missing historical records
- Lack of failure intelligence

Needs:

- Instant knowledge retrieval
 - Similar incident recommendations
 - Maintenance suggestions
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Quality Engineer

Responsibilities:

- Audits
- Compliance monitoring
- CAPA activities

Pain Points:

- Fragmented documentation
- Manual evidence collection

Needs:

- Compliance intelligence
 - Audit evidence generation
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Plant Manager

Responsibilities:

- Production reliability
- Asset availability

Pain Points:

- Repeated failures
- Knowledge silos

Needs:

- Decision support
 - Risk alerts
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Field Technician

Responsibilities:

- Equipment maintenance
- Inspections

Needs:

- Mobile access
 - Voice interaction
 - SOP guidance
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Reliability Engineer

Needs:

- Failure pattern analysis
 - Root cause support
 - Predictive maintenance
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8. Stakeholders

Primary Stakeholders

- Plant Managers
- Maintenance Teams
- Reliability Engineers
- Quality Engineers
- Safety Teams

Secondary Stakeholders

- Operations Team
 - Compliance Team
 - Management
 - Auditors
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9. Product Scope

In Scope

Document ingestion

OCR

Knowledge graph construction

RAG chatbot

Semantic search

Root Cause Analysis Agent

Maintenance Intelligence Agent

Lessons Learned Engine

Compliance Intelligence Agent

Hybrid Retrieval

Multi-agent architecture

Role-based access

Audit trail

Source citations

Confidence scoring

Out of Scope

SCADA Integration

Real-time sensor ingestion

IoT devices

ERP integrations

SAP integrations

Autonomous maintenance execution

Live PLC communication

Predictive control systems

10. Value Proposition

Traditional Systems

- Documents are isolated
- Search is keyword based
- Knowledge is hidden
- No reasoning capability

IntelliPlant AI

- Unified intelligence layer
- Semantic understanding
- Graph relationships
- Multi-agent reasoning
- Explainable AI
- Knowledge preservation

- Cross-functional insights
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Product Tagline

"Transform Documents into Industrial Intelligence."